Scientific Basis of the Serpentine Cold-Plate Thermo-Hydraulic Model

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Abstract

A detailed derivation of the reduced-order geometry, fluid-flow, and heat-transfer equations implemented in the browser-side solver.

Contents

The dashboard couples four calculation stages that execute in sequence: a geometric optimizer that lays out a serpentine (meandering) channel inside the plate footprint, a one-dimensional hydraulic model that sizes the flow rate and pressure drop, a lumped natural-convection balance that estimates how much of the chip's heat escapes to ambient air before reaching the water, and a two-dimensional finite-difference conduction solver that reconstructs the copper temperature field. Each stage is described below together with the exact expressions used in the code, and every symbol is given with its physical unit as implemented (SI units unless noted; geometric inputs are entered in millimetres and converted internally through the factor mm = 0.001 m/mm).

1 Channel-path synthesis and geometric optimization

Before any thermal quantity is evaluated, the model must generate a single continuous centerline that meanders back and forth across the plate a chosen number of times, called the number of passes N. For a candidate orientation (horizontal or vertical runs) the parallel-leg pitch, i.e. the center-to-center spacing between two adjacent straight channel legs, is set by the channel width and the wall thickness that separates neighboring legs:

$$p = w_{\text{ch}} + t_w. \tag{1}$$

p, leg pitch (center-to-center spacing between adjacent channel legs); unit: m.

w_{ch}, channel width (parameter channelW); unit: m.

t_w, wall thickness separating two adjacent legs (parameter wallT); unit: m.

The N leg centerlines are then distributed symmetrically about the plate centerline, offset by a vertical-shift term, and successive legs are joined by 180° circular return bends of radius equal to half the leg pitch:

$$R_{\text{bend}} = \frac{p}{2}, \tag{2}$$

$$x_i = -\frac{1}{2}\left[N w_{\text{ch}} + (N-1)t_w\right] + \frac{w_{\text{ch}}}{2} + ip + \Delta y, \qquad i = 0, \ldots, N-1. \tag{3}$$

R_{bend}, radius of the 180° return bend joining two consecutive legs; unit: m.

x_i, centerline offset of leg i measured across the meander direction; unit: m.

Δy, user-defined vertical shift of the whole pattern (parameter yShift); unit: m.

Because the exact number of legs, their lateral shift, the starting side, and the direction of traversal are all free choices, the algorithm performs a discrete search over N (within ±2 of the requested pass count), five candidate shift values, both starting sides and both traversal directions, and for the horizontal-or-vertical routing option. Each candidate is discarded if it fails to cover the required cooling zone with at least 99.9% of a 17×17 sampling grid lying within half a pitch of the centerline, and the remaining candidates are ranked with a weighted objective function that penalizes the length of the inlet/outlet connector segments, the total wetted length, the deviation from the requested pass count, and rewards routes that keep more of their length under the CPU footprint:

$$J = 8L_{\text{conn}} + L_{\text{path}} + 0.12|N - N_{\text{target}}| - 0.25 f_{\text{cpu}} - 0.10 m_{\text{cov}}. \tag{4}$$

J, route score to be minimized (dimensionless, mm-weighted); unit: mm.

L_{conn}, combined length of the inlet and outlet connector segments; unit: mm.

L_{path}, total centerline (arc) length of the candidate route; unit: mm.

N_{target}, requested number of passes (parameter numPasses); unit: 1.

f_{cpu}, fraction of path samples lying under the CPU footprint; unit: $1 (0-1)$.

m_{cov}, coverage safety margin, i.e. minimum clearance between the sampling grid and the centerline minus p/2; unit: mm.

The lowest-scoring candidate becomes the working geometry. Its cumulative arc length is tabulated at every polyline vertex through a running sum, which is the coordinate s used by every downstream one-dimensional profile (velocity, water temperature, local heat flux):

$$s_0 = 0, \tag{5}$$

$$s_k = s_{k-1} + \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2}. \tag{6}$$

s_k, cumulative curvilinear coordinate at vertex k; unit: $\text{mm(convertedtombeforeuseinthesolver)}$.

2 Inlet/outlet connector search, bend parametrization, and the coverage metric

Each candidate meander must still be linked to the fixed inlet and outlet port coordinates chosen by the user, and the two connecting segments are not part of the meander pattern itself, so they are solved as a small independent sub-problem for every candidate. For a given port point and the corresponding free end of the meander, three candidate connector shapes are generated: a single direct segment between the two points, an L-shaped path that turns first along the horizontal direction, and an L-shaped path that turns first along the vertical direction.

$$\mathcal{P}_1 = \{\mathbf{a}, \mathbf{b}\}, \qquad \mathcal{P}_2 = \{\mathbf{a}, (b_x, a_y), \mathbf{b}\}, \qquad \mathcal{P}_3 = \{\mathbf{a}, (a_x, b_y), \mathbf{b}\}. \tag{7}$$

a, fixed port coordinate (inlet or outlet); unit: mm.

b, free end of the meander core to be connected; unit: mm.

$P_1,\ P_2,\ P_3$, the three candidate connector polylines (direct, horizontal-first, vertical-first); unit: 1.

A candidate is rejected outright if any of its vertices would place the connector centerline closer than one port radius to the plate edge in either direction, since that would leave insufficient material for the port boss. Among the remaining feasible candidates the one with the smallest total length is kept:

$$\text{feasible}(\mathcal{P}) \iff \begin{cases} |x_k| \le \dfrac{L_p}{2} - \dfrac{d_{\text{port,OD}}}{2}, \\ begin equation 3pt]|y_k| \le \dfrac{W_p}{2} - \dfrac{d_{\text{port,OD}}}{2}, \end{cases} \qquad \text{for every vertex } k \text{ of } \mathcal{P}. \tag{8}$$

$$\mathcal{P}^* = \underset{\substack{\mathcal{P}\in\{\mathcal{P}_1,\mathcal{P}_2,\mathcal{P}_3\}\\ \text{feasible}(\mathcal{P})}}{\arg\min} \; L(\mathcal{P}), \tag{9}$$

$$L(\mathcal{P}) = \sum_k \sqrt{(x_{k+1}-x_k)^2 + (y_{k+1}-y_k)^2}. \tag{10}$$

P^*, selected connector polyline for one port; unit: 1.

$L(P)$, total length of a candidate polyline, summed segment by segment; unit: mm.

The two selected connectors are appended to the front and back of the meander core to form the single continuous path that is scored by the objective function J introduced above; if either port cannot be connected by any feasible candidate for a given N, shift, or orientation, that whole candidate route is discarded before scoring, which is why the optimizer occasionally reports that no route could be found for extreme port placements.

The 180° return bend joining two consecutive straight legs is not approximated by a sharp corner; it is built as an explicit semicircular arc so that the reported hydraulic length and the plotted velocity field reflect the real curvature the coolant follows. The arc center is placed at the midpoint between the two leg centerlines, its radius equals half the leg pitch so that the arc is exactly tangent to both straight legs at the hand-off points, and eighteen equally spaced angular stations are sampled across a 180° sweep whose starting angle depends on which side of the plate the meander is currently turning toward:

$$\mathbf{c}_{\text{bend}} = \left(x_{\text{leg}}, \frac{y_i + y_{i+1}}{2}\right), \qquad R_{\text{bend}} = \frac{p}{2}. \tag{11}$$

$$\begin{pmatrix} x(\theta) \\ y(\theta) \end{pmatrix} = \mathbf{c}_{\text{bend}} + R_{\text{bend}} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad \theta \in [\theta_0, \theta_0 + \pi], \tag{12}$$

with the arc sampled at 18 angular stations. $\mathbf{c}_{\text{bend}}$, center of the 180° return arc; unit: mm.

θ_0, starting angle of the sweep, ±π/2 depending on turn direction; unit: rad.

Whether a candidate meander actually cools the region the user asked it to cool is checked directly rather than inferred indirectly from the pitch alone. A regular 17×17 grid of sample points is generated over the rectangular cooling zone, and for every sample point the shortest distance to the piecewise-linear centerline is computed by projecting the point onto each segment of the path in turn and clamping the projection parameter to the segment's extent - the standard closed-form point-to-segment distance used in computational geometry:

$$t = \text{clamp}\left(\frac{(\mathbf{p}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})}{\|\mathbf{b}-\mathbf{a}\|^2}, 0, 1\right), \tag{13}$$

$$d(\mathbf{p}, \text{seg}) = \|\mathbf{p} - [\mathbf{a} + t(\mathbf{b}-\mathbf{a})]\|. \tag{14}$$

t, normalized projection of the sample point onto the segment, clamped to [0,1]; unit: 1.

$d(\mathbf{p}, \text{seg})$, perpendicular (or nearest end-point) distance from sample point p to the segment; unit: mm.

The minimum of d over every segment of the path gives the distance from that sample point to the channel centerline. A point is considered cooled if this minimum distance does not exceed half the leg pitch - i.e. it lies no farther from the centerline than the middle of the metal web separating two channel legs - and the coverage fraction is the count of cooled sample points divided by the total 17×17 = 289 points. The single largest distance found across all sample points is also retained and converted into a safety margin relative to the half-pitch threshold; this margin both disqualifies routes with cov below 99.9% and enters the objective function J and the results table as the coverage safety margin:

$$\mathrm{cov} = \frac{1}{289}\sum_{\mathbf{p}} \mathbf{1}\left[\min_{\mathrm{seg}} d(\mathbf{p}, \mathrm{seg}) \le \frac{p}{2}\right], \tag{15}$$

$$\mathrm{margin} = \frac{p}{2} - \max_{\mathbf{p}}\left(\min_{\mathrm{seg}} d(\mathbf{p}, \mathrm{seg})\right). \tag{16}$$

cov, fraction of the cooling-zone sampling grid within reach of the channel; unit: $-(0-1), \mathrm{route rejected if cov} < 0.999$.

m_{cov}, coverage safety margin, used as a rejection/ranking criterion; unit: mm.

3 Thermophysical properties of the surrounding air

The natural-convection and radiation losses from the plate to ambient require air properties evaluated at the mean film temperature between the plate surface and the ambient air. Density follows the ideal-gas law, viscosity follows Sutherland's law [6], thermal conductivity is a linear fit valid near room temperature, and the remaining transport groups follow directly from their definitions.

$$\rho_{\mathrm{air}} = \frac{p_{\mathrm{atm}}}{R_{\mathrm{air}} T}. \tag{17}$$

ρ_{air}, air density; unit: $\mathrm{kg/m^3}$.

p_{atm}, ambient pressure (parameter pressure); unit: Pa.

R_{air}, specific gas constant of air, 287.058; unit: J/(kg·K).

T, film (evaluation) temperature; unit: K.

$$\mu_{\mathrm{air}} = 1.716 \times 10^{-5}\left(\frac{T}{273.15}\right)^{3/2}\frac{273.15 + 111}{T + 111}. \tag{18}$$

μ_{air}, dynamic viscosity of air, Sutherland's formula (reference viscosity 1.716×10^{-5} Pa·s, Sutherland constant 111 K); unit: Pa·s.

$$k_{\mathrm{air}} = 0.0241 + 7.72 \times 10^{-5}(T - 273.15). \tag{19}$$

k_{air}, thermal conductivity of air, linear fit about 0°C; unit: W/(m·K).

$$c_{p,\mathrm{air}} = 1007, \qquad \mathrm{Pr} = \frac{c_{p,\mathrm{air}}\mu_{\mathrm{air}}}{k_{\mathrm{air}}}, \tag{20}$$

$$\nu = \frac{\mu_{\mathrm{air}}}{\rho_{\mathrm{air}}}, \qquad \alpha = \frac{k_{\mathrm{air}}}{\rho_{\mathrm{air}} c_{p,\mathrm{air}}}, \qquad \beta = \frac{1}{T}. \tag{21}$$

$c_{p,\mathrm{air}}$, specific heat of air at constant pressure (constant); unit: J/(kg·K).

Pr, Prandtl number of air; unit: 1.

ν, kinematic viscosity of air; unit: m²/s.

α, thermal diffusivity of air; unit: m²/s.

β, volumetric thermal-expansion coefficient of an ideal gas; unit: 1/K.

4 Coolant flow rate and internal duct hydraulics

The mass flow rate of water is not a free input; it is derived from the energy balance that is required to keep the bulk water-temperature rise below the user's target while carrying the portion of the chip heat that actually enters the coolant. The channel cross-section is treated as a smooth rectangular duct of width equal to the channel width and height equal to the channel height.

$$\dot{m} = \frac{Q}{c_{p,w}\Delta T_{\text{target}}}, \qquad \dot{V} = \frac{\dot{m}}{\rho_w}. \tag{22}$$

$\dot{m}$, coolant mass flow rate; unit: kg/s.

Q, total chip heat load (parameter heatLoad); unit: W.

$c_{p,w}$, water specific heat (parameter waterCp); unit: J/(kg·K).

ΔT_{target}, target water temperature rise (parameter maxRise); unit: K.

$\dot{V}$, volumetric flow rate; unit: m³/s.

ρ_w, water density (parameter waterRho); unit: kg/m³.

$$A_c = w_{\text{ch}} H_{\text{ch}}, \qquad D_h = \frac{2 w_{\text{ch}} H_{\text{ch}}}{w_{\text{ch}} + H_{\text{ch}}}, \tag{23}$$

$$U = \frac{\dot{V}}{A_c}, \qquad \text{Re} = \frac{\rho_w U D_h}{\mu_w}. \tag{24}$$

A_c, channel cross-sectional flow area; unit: m².

H_{ch}, channel height (parameter channelH); unit: m.

D_h, hydraulic diameter of the rectangular duct; unit: m.

U, mean coolant velocity; unit: m/s.

Re, Reynolds number of the internal flow; unit: 1.

μ_w, water dynamic viscosity (parameter waterMu); unit: Pa·s.

The friction factor and the internal convective Nusselt number [12] are obtained from correlations for flow between parallel or rectangular boundaries, selected automatically according to the regime indicated by the Reynolds number and the channel aspect ratio.

$$\beta_{\text{AR}} = \frac{\min(w_{\text{ch}}, H_{\text{ch}})}{\max(w_{\text{ch}}, H_{\text{ch}})}. \tag{25}$$

β_{AR}, channel cross-section aspect ratio; unit: $1(0-1)$.

For laminar flow (Re < 2300) the fully-developed friction factor and Nusselt number [12] follow the fifth-order Shah-London [2] polynomial fits for rectangular ducts with uniform wall heat flux:

$$\mathrm{Po}(\beta_{\mathrm{AR}}) = 96\Big(1 - 1.3553\beta_{\mathrm{AR}} + 1.9467\beta_{\mathrm{AR}}^2 - 1.7012\beta_{\mathrm{AR}}^3 \tag{26}$$

$$+ 0.9564\beta_{\mathrm{AR}}^4 - 0.2537\beta_{\mathrm{AR}}^5\Big), \tag{27}$$

$$f = \frac{\mathrm{Po}}{\mathrm{Re}}. \tag{28}$$

$$\mathrm{Nu} = 8.235\Big(1 - 2.0421\beta_{\mathrm{AR}} + 3.0853\beta_{\mathrm{AR}}^2 - 2.4765\beta_{\mathrm{AR}}^3 + 1.0578\beta_{\mathrm{AR}}^4 - 0.1861\beta_{\mathrm{AR}}^5\Big). \tag{29}$$

Po, Poiseuille number of the laminar rectangular duct; unit: 1.

f, Darcy friction factor; unit: 1.

Nu, internal Nusselt number [12] (constant heat-flux boundary condition); unit: 1.

For turbulent flow (Re ≥ 2300) the friction factor follows the Haaland explicit approximation [4] to the Colebrook equation for a hydraulically smooth duct, and the Nusselt number [12] follows the Gnielinski correlation [3], floored at the laminar plateau value of 3.66:

$$f = \left[-1.8\log_{10}\left(\frac{6.9}{\mathrm{Re}}\right)\right]^{-2}. \tag{30}$$

$$\mathrm{Nu} = \max\left\{\frac{(f/8)(\mathrm{Re} - 1000)\mathrm{Pr}}{1 + 12.7\sqrt{f/8}\left(\mathrm{Pr}^{2/3} - 1\right)},\ 3.66\right\}. \tag{31}$$

The average internal convective heat-transfer coefficient used everywhere downstream is then recovered from the definition of the Nusselt number [12]:

$$h_{\mathrm{int}} = \frac{\mathrm{Nu}\, k_w}{D_h}. \tag{32}$$

h_{int}, average internal convective heat-transfer coefficient; unit: W/(m²·K).

k_w, water thermal conductivity (parameter waterK); unit: W/(m·K).

5 Hydraulic pressure drop

The total pressure drop combines a distributed (Darcy-Weisbach [1]) friction term evaluated over the full wetted path length with a minor-loss term that lumps every 180° return bend and the inlet/outlet port transitions into loss coefficients applied to the dynamic pressure:

$$q_{\mathrm{dyn}} = \frac{1}{2}\rho_w U^2. \tag{33}$$

$$\Delta p = f\frac{L_{\mathrm{hyd}}}{D_h} q_{\mathrm{dyn}} + \left(n_{\mathrm{bend}} K_{\mathrm{bend}} + K_{\mathrm{ports}}\right) q_{\mathrm{dyn}}. \tag{34}$$

q_{dyn}, dynamic pressure of the coolant; unit: Pa.

Δp, total predicted pressure drop; unit: Pa.

L_{hyd}, total wetted centerline length of the routed channel; unit: m.

n_{bend}, number of 180° return bends, equal to (N−1)+2; unit: 1.

K_{bend}, minor-loss coefficient per 180° bend (parameter Kbend); unit: 1.

K_{ports}, combined minor-loss coefficient of the inlet/outlet ports (parameter Kports); unit: 1.

The result is compared against the maximum allowable pressure drop entered by the user; the design is flagged PASS when Δp does not exceed that ceiling and FAIL otherwise.

6 Equilibrium plate surface temperature and natural-convection loss

Not all of the chip heat is captured by the coolant: some escapes directly from the exposed copper surfaces to the surrounding air by natural (buoyancy-driven) convection and, if enabled, by radiation. Because the natural-convection coefficients themselves depend on the unknown surface temperature, the model solves this sub-problem by fixed-point (successive-substitution) iteration on an assumed uniform plate surface temperature T_s, evaluating air properties at the film temperature $T_f^{\text{air}} = (T_s + T_{\text{amb}})/2$ at every iteration.

For the top face, the characteristic length used in the Rayleigh number [11] is the ratio of plate area to plate perimeter (the classical choice for horizontal-plate free-convection correlations), and the correlation used for the Nusselt number [12] depends on whether the finned side faces up or down:

$$L_c = \frac{L_p W_p}{2(L_p + W_p)}, \tag{35}$$

$$\text{Ra}_{\text{top}} = \frac{g\beta |T_s - T_{\text{amb}}| L_c^3}{\nu\alpha}. \tag{36}$$

L_c, characteristic length of the top face (area / perimeter); unit: m.

L_p, W_p, plate length and width (parameters plateL, plateW); unit: m.

g, gravitational acceleration, 9.81; unit: m/s².

Ra_{top}, Rayleigh number [11] of the top-face buoyant layer; unit: 1.

T_s, assumed uniform plate surface temperature (iteration variable); unit: K.

T_{amb}, ambient air temperature (parameter Tamb); unit: K.

$$\text{Nu}_{\text{top}} = \begin{cases} 0.54\text{Ra}_{\text{top}}^{1/4}, & \text{Ra}_{\text{top}} < 10^7, \text{ hot face upward,} \\ 0.15\text{Ra}_{\text{top}}^{1/3}, & \text{Ra}_{\text{top}} \geq 10^7, \text{ hot face upward,} \\ 0.27\text{Ra}_{\text{top}}^{1/4}, & \text{hot face downward.} \end{cases} \tag{37}$$

$$h_{\text{top}} = \frac{\text{Nu}_{\text{top}} k_{\text{air}}}{L_c}. \tag{38}$$

h_{top}, top-face natural-convection coefficient; unit: W/(m²·K).

For the four side faces, treated together as a single vertical plate of height equal to the plate height, the Churchill-Chu correlation [5] is used because it is valid across the entire laminar-to-turbulent range of Rayleigh number [11]:

$$\mathrm{Ra}_{\text{side}} = \frac{g\beta|T_s - T_{\text{amb}}|H_p^3}{\nu\alpha}. \tag{39}$$

$$\mathrm{Nu}_{\text{side}} = \left[0.825 + \frac{0.387\mathrm{Ra}_{\text{side}}^{1/6}}{\left[1 + (0.492/\mathrm{Pr})^{9/16}\right]^{8/27}}\right]^2. \tag{40}$$

$$h_{\text{side}} = \frac{\mathrm{Nu}_{\text{side}} k_{\text{air}}}{H_p}. \tag{41}$$

H_p, plate height (parameter plateH); unit: m.

h_{side}, side-face natural-convection coefficient; unit: W/(m²·K).

When radiation is enabled, a linearized radiative coefficient is added, obtained by factoring the fourth-power Stefan-Boltzmann difference into a coefficient multiplying the same linear temperature difference used for the convective terms:

$$h_{\text{rad}} = \varepsilon\sigma(T_s + T_{\text{amb}})(T_s^2 + T_{\text{amb}}^2). \tag{42}$$

h_{rad}, linearized radiative heat-transfer coefficient; unit: W/(m²·K).

ε, copper surface emissivity (parameter emissivity); unit: $1(0-1)$.

σ, Stefan-Boltzmann constant [7], 5.670374419×10^{-8}; unit: W/(m²·K⁴).

The three coefficients are combined with the corresponding exposed areas (the top area excludes the two port openings; the side area is the plate perimeter times its height) into an overall external conductance, which is then used to estimate how much heat is rejected directly to ambient and, by difference, how much must be absorbed by the coolant:

$$A_{\text{top}} = L_pW_p - 2\pi\left(\frac{d_{\text{port,OD}}}{2}\right)^2, \tag{43}$$

$$A_{\text{side}} = 2(L_p + W_p)H_p. \tag{44}$$

$$UA = h_{\text{top}}A_{\text{top}} + h_{\text{side}}A_{\text{side}} + h_{\text{rad}}(A_{\text{top}} + A_{\text{side}}). \tag{45}$$

$$Q_{\text{ext}} = UA\max(T_s - T_{\text{amb}}, 0), \tag{46}$$

$$Q_w = \max(Q - Q_{\text{ext}}, 0.05Q). \tag{47}$$

A_{top}, A_{side}, effective top and side heat-loss areas; unit: m².

$d_{\text{port,OD}}$, port outer diameter (parameter portOD); unit: m.

UA, overall external thermal conductance; unit: W/K.

Q_{ext}, heat rejected directly to ambient air; unit: W.

Q_w, heat delivered into the coolant, floored at 5% of Q; unit: W.

The surface temperature is then updated from a simple two-resistance model that lumps the internal convective resistance between the channel wall and the bulk coolant with the conduction resistance of the base plate under the CPU footprint, driven by the heat presently assigned to the water and referenced to the coolant's mean bulk temperature:

$$R_{\text{int}} = \frac{1}{h_{\text{int}} A_{\text{int}}}, \qquad A_{\text{int}} = (w_{\text{ch}} + 2H_{\text{ch}})L_{\text{hyd}}. \tag{48}$$

$$R_{\text{cond}} = \frac{t_{\text{base}}}{k_{\text{Cu}} L_{\text{cpu}} W_{\text{cpu}}}. \tag{49}$$

$$T_s^{\text{new}} = \frac{1}{2}\left[T_{\text{in}} + (T_{\text{in}} + \Delta T_{\text{target}})\right] + Q_w(R_{\text{int}} + R_{\text{cond}}). \tag{50}$$

R_{int}, internal convective thermal resistance between channel wetted wall and coolant; unit: K/W.

A_{int}, internal wetted area of the channel (three wetted sides per unit length); unit: m^2.

R_{cond}, conduction resistance of the base plate directly under the CPU footprint; unit: K/W.

t_{base}, base-plate thickness under the channel (parameter baseT); unit: m.

k_{Cu}, copper thermal conductivity (parameter copperK); unit: W/(m·K).

L_{cpu}, W_{cpu}, CPU die length and width (parameters cpuL, cpuW); unit: m.

T_{in}, coolant inlet temperature (parameter Tin); unit: K.

The updated surface temperature is blended with the previous iterate using a fixed under-relaxation factor of 0.6/0.4 and the loop repeats - recomputing air properties, Rayleigh number [11]s, and the conductance - until $|T_s^{(n+1)} - T_s^{(n)}| < 10^{-4}$ K or the iteration budget (parameter maxIterNat) is exhausted. Once converged, the natural-convection loss is capped at 35% of the total chip power, and the water-side heat load is finalized by difference:

$$Q_{\text{nat}} = \min\left[UA \max(T_s - T_{\text{amb}}, 0), 0.35Q\right], \tag{51}$$

$$Q_{\text{water}} = Q - Q_{\text{nat}}. \tag{52}$$

Q_{nat}, final converged heat loss to ambient air; unit: W.

Q_{water}, final heat load absorbed by the coolant; unit: W.

7 Distribution of heat pickup and coolant temperature rise along the channel

Heat does not enter the coolant uniformly along the serpentine path: the sections of channel that pass beneath the CPU footprint pick up heat preferentially, while sections outside that footprint still receive a small background fraction representing lateral spreading through the copper. A weighting function is therefore assigned at every vertex of the path, integrated over the path length, and used to normalize the local linear heat-generation rate so that its integral reproduces the total water-side heat load:

$$\omega(s) = 0.12 + 0.88\,\delta_{\text{cpu}}(s), \qquad \delta_{\text{cpu}}(s) = \begin{cases} 1, & s \text{ lies under the CPU footprint,} \\ 0, & \text{otherwise.} \end{cases} \tag{53}$$

$$I = \int_0^{L_{\text{hyd}}} \omega(s)\,\mathrm{d}s, \tag{54}$$

where the integral is evaluated by the composite trapezoidal rule.

$$q'(s) = \frac{Q_{\text{water}}\omega(s)}{I}. \tag{55}$$

$\omega(s)$, local heat-pickup weighting factor along the path; unit: 1.

I, path integral of the weighting function; unit: m.

$q'(s)$, local linear heat-generation rate entering the coolant per unit channel length; unit: W/m.

The bulk coolant temperature is obtained by integrating the local energy balance $\mathrm{d}T_f/\mathrm{d}s = q'(s)/(\dot{m}c_{p,w})$ along the path using the trapezoidal rule. The cumulative absorbed heat is tabulated in parallel:

$$T_f(s_k) = T_f(s_{k-1}) + \frac{\frac{1}{2}\left[q'(s_k) + q'(s_{k-1})\right](s_k - s_{k-1})}{\dot{m}c_{p,w}}. \tag{56}$$

$$Q_{\text{cum}}(s_k) = Q_{\text{cum}}(s_{k-1}) + \frac{1}{2}\left[q'(s_k) + q'(s_{k-1})\right](s_k - s_{k-1}). \tag{57}$$

$T_f(s)$, bulk coolant temperature as a function of position along the channel; unit: K.

$Q_{\text{cum}}(s)$, cumulative heat absorbed by the coolant from the inlet up to position s; unit: W.

The value of T_f at the last vertex is reported as the coolant outlet temperature, and the local velocity plotted along the path is the mean channel velocity U modulated by a small empirical acceleration factor applied wherever the local direction change between successive segments exceeds a threshold, representing the flow acceleration through a bend.

8 Two-dimensional steady conduction field in the copper base

The copper base and lid are represented by a single effective two-dimensional conducting layer discretized on an $N_x \times N_y$ grid. Each control area exchanges heat by in-plane conduction, receives the CPU source where applicable, rejects heat to the local coolant where the channel is present, and loses heat to ambient through the top-surface convection and radiation terms.

The reduced-order conducting thickness is defined as

$$t_{\text{eff}} = t_{\text{base}} + \eta_h H_{\text{ch}}, \qquad \eta_h = 0.35. \tag{58}$$

Here, t_{base} is the physical copper thickness beneath the channel, H_{ch} is the channel height, and η_h is an empirical spreading factor. Thus, t_{eff} is not a second geometric thickness; it is an equivalent in-plane conduction thickness that accounts approximately for heat spreading through part of the channel-height region.

To avoid confusing the channel height H_{ch} with a heat-transfer coefficient, the areal coupling terms are denoted by calligraphic symbols:

$$q''_{\text{src}}(x, y) = \frac{Q}{L_{\text{cpu}} W_{\text{cpu}}}\, \chi_{\text{cpu}}(x, y), \tag{59}$$

$$\mathcal{H}_{\text{ch}}(x, y) = h_{\text{int}} \frac{w_{\text{ch}} + 2H_{\text{ch}}}{w_{\text{ch}}}\, \chi_{\text{ch}}(x, y), \tag{60}$$

$$\mathcal{H}_{\text{ex}} = h_{\text{top}} + h_{\text{rad}}. \tag{61}$$

The indicator χ_{cpu} equals one beneath the CPU footprint and zero elsewhere, while χ_{ch} equals one on channel-covered grid cells and zero elsewhere. Both $\mathcal{H}_{\text{ch}}$ and $\mathcal{H}_{\text{ex}}$ have units of $\mathrm{W\,m^{-2}\,K^{-1}}$.

The steady thin-plate energy equation is therefore

$$k_{\text{Cu}} t_{\text{eff}} \nabla_{\parallel}^2 T + q''_{\text{src}} - \mathcal{H}_{\text{ch}}(T - T_f) - \mathcal{H}_{\text{ex}}(T - T_{\text{amb}}) = 0, \tag{62}$$

where $\nabla_{\parallel}^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the in-plane Laplacian. Every term in Eq. (62) has units of heat flux, $\mathrm{W\,m^{-2}}$.

For a control area $\Delta x\, \Delta y$ centered at node (i, j), conservation of energy gives

$$\begin{aligned} k_{\text{Cu}} t_{\text{eff}} \Delta y & \left(\frac{T_{i+1,j} - T_{i,j}}{\Delta x} - \frac{T_{i,j} - T_{i-1,j}}{\Delta x} \right) \\ & + k_{\text{Cu}} t_{\text{eff}} \Delta x \left(\frac{T_{i,j+1} - T_{i,j}}{\Delta y} - \frac{T_{i,j} - T_{i,j-1}}{\Delta y} \right) \\ & + q''_{\text{src},i,j} \Delta x \Delta y - \mathcal{H}_{\text{ch},i,j}(T_{i,j} - T_{f,i,j}) \Delta x \Delta y - \mathcal{H}_{\text{ex}}(T_{i,j} - T_{\text{amb}}) \Delta x \Delta y = 0. \end{aligned} \tag{63}$$

After division by $\Delta x\, \Delta y$, define

$$K_x = \frac{k_{\text{Cu}} t_{\text{eff}}}{\Delta x^2}, \qquad K_y = \frac{k_{\text{Cu}} t_{\text{eff}}}{\Delta y^2}, \tag{64}$$

where K_x and K_y have units of $\mathrm{W\,m^{-2}\,K^{-1}}$. The Gauss–Seidel estimate becomes

$$T_{i,j}^{\text{GS}} = \frac{K_x(T_{i+1,j} + T_{i-1,j}) + K_y(T_{i,j+1} + T_{i,j-1}) + q''_{\text{src},i,j} + \mathcal{H}_{\text{ch},i,j} T_{f,i,j} + \mathcal{H}_{\text{ex}} T_{\text{amb}}}{2K_x + 2K_y + \mathcal{H}_{\text{ch},i,j} + \mathcal{H}_{\text{ex}}}. \tag{65}$$

Successive over-relaxation updates the node as

$$T_{i,j}^{(n+1)} = (1 - \omega_{\text{SOR}}) T_{i,j}^{(n)} + \omega_{\text{SOR}} T_{i,j}^{\text{GS}}, \qquad 0 < \omega_{\text{SOR}} < 2. \tag{66}$$

In the implementation, $\omega_{\text{SOR}} = \min(1.8, \text{relaxation})$. The quantities $q''_{\text{src},i,j}$ and $\mathcal{H}_{\text{ch},i,j}$ are zero outside the CPU and channel masks, respectively, and $T_{f,i,j}$ is interpolated from the nearest position on the one-dimensional coolant profile.

The four grid edges are treated as adiabatic (zero-gradient / mirror) boundaries by copying the adjacent interior row or column into the boundary row or column at the start of every sweep, which is the discrete equivalent of the Neumann condition ∂T/∂n = 0 at the plate perimeter; this is a deliberate simplification that keeps the in-plane conduction problem self-contained, given that the side-face heat loss already computed in the natural-convection loop (h_{side}) is applied only through the lumped external-loss budget UA rather than resolved locally at the plate edge, so it is not double-counted at the boundary nodes of the two-dimensional grid.

$$T_{0,j} = T_{1,j}, \quad T_{N_x-1,j} = T_{N_x-2,j}, \quad T_{i,0} = T_{i,1}, \quad T_{i,N_y-1} = T_{i,N_y-2}. \tag{67}$$

Every sweep visits the interior nodes row by row, computing the maximum absolute nodal change observed during that sweep as a residual, and the iteration stops as soon as that residual falls below the convergence tolerance or the iteration budget is exhausted, whichever occurs first:

$$\varepsilon^{(n)} = \max_{i,j} \left| T_{i,j}^{(n)} - T_{i,j}^{(n-1)} \right|, \qquad \text{stop if } \varepsilon^{(n)} < \text{tol or } n \geq n_{\max}. \tag{68}$$

$\varepsilon^{(n)}$, maximum nodal temperature change during sweep n, used as the convergence residual; unit: K.

tol, convergence tolerance (parameter tolerance, default 10^{-5}); unit: K.

$n_{\max}$, maximum number of sweeps allowed (parameter maxIterTemp, default 1600); unit: 1.

Once the sweep loop terminates, the maximum nodal temperature found among all grid nodes lying under the CPU footprint is reported as the peak CPU-region temperature - the single most safety-critical output of the whole model, since it is what is finally checked against the user's maximum-allowable chip temperature:

$$T_{\text{cpu}} = \max \{ T_{i,j} : \ (i,j) \text{ lies under the CPU footprint} \}. \tag{69}$$

9 Governing assumptions and known simplifications

As with any reduced-order engineering model intended to run instantly in a browser rather than in a full computational-fluid-dynamics package, several simplifying assumptions trade some physical fidelity for speed and robustness, and they are worth stating explicitly so that the numbers produced by the dashboard are interpreted with the right degree of confidence.

The plate surface temperature used in the natural-convection loop is a single lumped value rather than a spatially resolved field, so the model does not capture a locally hotter patch of copper directly above the CPU when estimating the air-side loss; it implicitly assumes the top and side natural-convection coefficients are representative of the whole exposed area. The internal flow correlations assume fully developed, hydrodynamically and thermally established flow along a straight duct of constant cross-section, so entrance-length effects after each 180° bend, and the local enhancement of heat transfer and pressure loss that a real bend produces beyond the lumped K_{bend} coefficient, are only approximated. The transition between the laminar and turbulent correlation sets is treated as a sharp switch at Re $= 2300$ rather than a smooth blend through the transitional regime, which can produce a small discontinuity in the reported friction factor and Nusselt number [12] for designs operating close to that Reynolds number. Axial conduction of heat along the water and along the copper in the flow direction is neglected in the one-dimensional energy balance that produces $T_f(s)$; only convective transport along s and local pickup from $q'(s)$ are retained. The radiative heat-transfer coefficient is a linearization of the Stefan-Boltzmann law that is exact only in the limit where it is evaluated with the correct T_s and T_{amb}, which the iterative loop does self-consistently, but it still assumes a single ambient radiative background temperature rather than a detailed view-factor calculation to real surrounding surfaces. Finally, the two-dimensional conduction solve and the one-dimensional hydraulic/thermal stages are only loosely coupled: the external loss coefficients ($h_{\text{top}}, h_{\text{rad}}$) and the coolant temperature profile $T_f(s)$ are computed once, upstream, and then held fixed while the SOR sweep converges the copper field, rather than iterating the two stages against each other to full mutual consistency.

10 Reported performance metrics and pass/fail evaluation

Once the hydraulic and thermal stages converge, the dashboard tabulates the derived performance quantities that characterize the design and compares the two governing constraints - pressure drop and peak chip temperature - against the user-specified limits to issue a PASS/FAIL verdict for each:

$$\dot{V}_{[\mathrm{L/min}]} = 60000\dot{V}, \qquad \Delta p \le \Delta p_{\max} \Rightarrow \mathrm{PASS}, \qquad T_{\mathrm{cpu}} \le T_{\mathrm{cpu,max}} \Rightarrow \mathrm{PASS}. \tag{70}$$

$\Delta p_{\max}$, maximum allowable pressure drop (parameter maxDP); unit: kPa.

$T_{\mathrm{cpu,max}}$, maximum allowable CPU temperature (parameter maxCpu); unit: °C.

Together, these stages form a closed, self-consistent reduced-order model: the geometric optimizer fixes the flow path and its length, the hydraulic correlations fix the flow regime and pressure loss for a coolant flow rate sized from the target temperature rise, the natural-convection fixed-point loop partitions the chip's heat between air and water, the one-dimensional energy balance propagates the resulting water-side heat load into a bulk coolant temperature profile along the channel, and the two-dimensional finite-difference conduction solve translates that coolant profile, together with the direct CPU heat flux, into a full copper temperature field from which the worst-case junction-adjacent temperature is extracted.

11 Numerical-method summary: integration rules, iteration budgets, and default tolerances

Two distinct families of numerical method are used, and it is useful to state their formal order of accuracy and the default control parameters exposed in the fluid/solver input tab, since these govern how closely the reduced-order results track the underlying continuous equations. Every line integral along the channel path - the arc-length coordinate s, the coverage-weight integral I, the coolant temperature profile T_f(s), and the cumulative heat Q_cum(s) - is evaluated with the composite trapezoidal rule applied between consecutive polyline vertices, which is exact for a piecewise-linear integrand and has a local truncation error of order $O(\Delta s^2)$ on smoothly varying integrands, so refining the points-per-straight and points-per-bend settings (parameters pathStraight and pathBend) tightens these profiles directly. The two-dimensional copper-temperature field is solved with the finite-difference/SOR scheme described above, whose spatial truncation error is $O(\Delta x^2+\Delta y^2)$ and whose iterative (not spatial) error is controlled independently by the tolerance and maxIterTemp settings; because these are two different error sources, tightening the SOR tolerance alone cannot compensate for a coarse grid, and refining the grid (gridNx, gridNy) alone cannot compensate for stopping the sweep early.

$$\mathcal{O}(\Delta s^2), \qquad \mathcal{O}(\Delta x^2 + \Delta y^2), \qquad e_{\mathrm{iter}} \approx \frac{\varepsilon^{(n)}}{1-\rho_{\mathrm{SOR}}}. \tag{71}$$

ρ_{SOR}, spectral radius of the SOR iteration matrix (not computed explicitly by the solver, governs how quickly $\varepsilon^{(n)}$ decays); unit: 1.

The natural-convection equilibrium loop uses simple successive substitution with fixed under-relaxation (weights 0.6 old / 0.4 new) rather than a Newton-type method, trading a guaranteed quadratic convergence rate for a scheme that cannot diverge even when the $UA(T_s)$ relationship is strongly nonlinear at high Rayleigh number [11]; it is capped at maxIterNat sweeps (default 50) and a fixed absolute tolerance of 10^{-4} K on T_s. All three iterative processes - the geometric search (a finite discrete enumeration, not an iteration in the numerical-analysis sense), the natural-convection loop, and the SOR sweep - are run to their own independent convergence criteria before the next stage begins, which is what allows the browser to complete a full evaluation, typically several hundred thousand floating-point operations, within a fraction of a second on a single UI thread.

12 Nomenclature

The table below cross-references every symbol used in this document with the corresponding variable name in the source code, a short description, and its unit as used internally by the solver (SI units throughout, independent of the millimetre/°C/kPa units shown in the input fields, which are converted at the point of use).

Symbol	Code variable	Description	Unit
p	channelW + wallT	leg pitch between adjacent channel legs	m
w_{ch}	channelW	channel width	m
H_{ch}	channelH	channel height	m
t_w	wallT	wall thickness between adjacent legs	m
N	numPasses (optimized)	number of meander passes	1
Δy	yShift	vertical shift of the meander pattern	m
R_{bend}	p/2 (derived)	180° return-bend radius	m
J	score (derived)	route objective score	mm
cov	coverage[0] (derived)	cooling-zone coverage fraction	1
m_{cov}	coverage[1] (derived)	coverage safety margin	mm
s	arc length s (derived)	curvilinear path coordinate	m
ρ_{air}	air.rho (derived)	air density	kg/m³
μ_{air}	air.mu (derived)	air dynamic viscosity	Pa·s
k_{air}	air.k (derived)	air thermal conductivity	W/(m·K)
Pr	air.Pr / Pr (derived)	Prandtl number	1
$\dot{m}$	mdot (derived)	coolant mass flow rate	kg/s
$\dot{V}$	vflow (derived)	coolant volumetric flow rate	m³/s
U	U (derived)	mean coolant velocity	m/s
D_h	Dh (derived)	hydraulic diameter	m
Re	Re (derived)	Reynolds number	1
β_{AR}	beta (derived)	channel aspect ratio	1
f	f (derived)	Darcy friction factor	1
Nu	Nu (derived)	internal Nusselt number	1
h_{int}	hAvg	internal convective coefficient	W/(m²·K)
Δp	dp	total pressure drop	Pa

Symbol	Code variable	Description	Unit
K_{bend}	Kbend	per-bend minor-loss coefficient	1
K_{ports}	Kports	port minor-loss coefficient	1
T_s	Ts	lumped plate surface temperature	K
h_{top}	hTop	top-face convection coefficient	W/(m²·K)
h_{side}	hSide	side-face convection coefficient	W/(m²·K)
h_{rad}	hRad	linearized radiation coefficient	W/(m²·K)
ε	emissivity	copper surface emissivity	1
UA	UA	overall external conductance	W/K
Q_{ext}	Qext / Qnat	heat lost to ambient air	W
$Q_w,\ Q_{\text{water}}$	Qw / Qwater	heat delivered to coolant	W
R_{int}	Rint	internal convective resistance	K/W
R_{cond}	Rcond	base conduction resistance	K/W
$\omega(s)$	weights (derived)	local heat-pickup weighting	1
$q'(s)$	qprime (derived)	local linear heat-generation rate	W/m
$T_f(s)$	Tf (derived)	bulk coolant temperature profile	K
$Q_{\text{cum}}(s)$	cum (derived)	cumulative absorbed heat	W
t_{eff}	tEff (derived)	equivalent in-plane conduction thickness, Eq. (58)	m
k_{Cu}	copperK	copper thermal conductivity	W/(m·K)
K_x, K_y	Kx, Ky (derived)	finite-difference areal conduction coefficients	W/(m^2 K)
ω_{SOR}	relaxation (clipped)	SOR relaxation factor	1
T_{cpu}	Tcpu (derived)	peak CPU-region temperature	K
$\Delta p_{\max}$	maxDP	maximum allowable pressure drop	kPa
$T_{\text{cpu,max}}$	maxCpu	maximum allowable CPU temperature	°C

13 References

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